

EXCERCISE 1

--1.1

--1

```
SELECT * FROM dba_data_files;
```

/*It shows a list of files stored C:\ORACLE19C\ORADATA\ORCL\ with files with the extension .DBF*/

--NOTE Directory location depend on installation folder

-- DBF means database files

--Database table's physical data or raw data are stored in DBF or as a DBF.

--2

-- Created after finding the list of data files location and created folder by the use of windows explorer

--3

```
CREATE TABLESPACE CURRENTTB
```

```
DATAFILE 'C:\ORACLE19C\ORADATA\ORCL\CURRENT\CURRENTTB01.DBF' SIZE 32M
```

```
AUTOEXTEND ON;
```

-- Note : Pay close attention to the directory of desired creation

-- Use '.DBF' for the file extension at the end of the database filename

EXCERCISE 2

--2.1

```
CREATE TABLE BigData ( ID int, GUID char(100)) TABLESPACE CURRENTTB;
```

--2.2

/*Turned off auto extended so the when the records total size passes 32M in this case shows an error once the tablespace database datafile size is passed gives an error or notification*/

```
ALTER DATABASE DATAFILE 'C:\ORACLE19C\ORADATA\ORCL\CURRENT\CURRENTTB01.DBF' AUTOEXTEND OFF;
```

-- As the word says 'AUTOEXTENDED' it increase the database datafile size to adjust with the records inputted sizes

```
DECLARE
```

```
I NUMBER;
```

```
BEGIN
```

```
I := 0;
```

```
WHILE (TRUE)
```

```
LOOP
```

```
INSERT INTO bigdata VALUES (I, SYS_GUID());
```

```
COMMIT;
```

```
I := i + 1;
```

```
END LOOP;
```

```
END;
```

```
--2.3
```

```
ALTER DATABASE DATAFILE 'C:\ORACLE19C\ORADATA\ORCL\CURRENT\CURRENTTB01.DBF' RESIZE  
60M;
```

```
--2.4
```

```
SELECT COUNT(*) "Total Records" FROM bigdata;  
/*So it maxed at a populated total of 27286 records at the initial datafile size of  
32M with AUTOEXTEND OFF*/
```

```
INSERT INTO bigdata VALUES (27287, SYS_GUID());
```

```
/*Record was inserted successfully since the database datafile for the tablespace has  
been increase to accommodate more  
record with the extra 28M resize difference */
```

```
--2.5
```

```
ALTER TABLESPACE CURRENTTB  
ADD DATAFILE 'C:\ORACLE19C\ORADATA\ORCL\CURRENT\CURRENTTB02.DBF'  
SIZE 50M AUTOEXTEND OFF;
```

```
--2.6
```

```
DESC DBA_DATA_FILES  
DESC DBA_EXTENTS  
DESC DBA_FREE_SPACE
```

```
-- Select * from
```

```
SELECT DF.TABLESPACE_NAME, DF.FILE_NAME , ROUND(DF.BYTES/1024/1024) Total_Size_Mb,  
NVL(ROUND(FreeBytes/1024/1024), 0) Free_Space_Mb , NVL(ROUND(UsedBytes/1024/1024), 0)  
Used_Space_Mb  
FROM DBA_DATA_FILES DF  
LEFT JOIN (  
    SELECT File_Id, SUM(BYTES) UsedBytes  
    from dba_extents  
    group by file_id  
    ) ext on df.file_id = ext.file_id  
LEFT JOIN (  
    SELECT File_Id, SUM(bytes) FreeBytes  
    from dba_free_space  
    group by file_id  
    ) free on df.file_id = free.file_id  
WHERE DF.TABLESPACE_NAME = 'CURRENTTB';
```

EXCERCISE 3

--3.1

```
CREATE TABLESPACE ARCHIVETB
DATAFILE 'C:\ORACLE19C\ORADATA\ORCL\ARCHIVE\ARCHIVETB01.DBF' SIZE 32M
AUTOEXTEND ON;
```

```
CREATE TABLE BigDataHist ( ID int, GUID char(1000)) TABLESPACE ARCHIVETB;
```

--3.2

```
INSERT INTO BigDataHist (ID, GUID)
SELECT bd.ID, bd.GUID FROM bigdata bd
WHERE ROWNUM < 11;
```

```
SELECT * FROM BigDataHist;
```

--3.3

```
ALTER TABLESPACE ARCHIVETB READ ONLY;
```

--3.4

```
SELECT * FROM BigDataHist;
```

/*Yes I was able to perform a select statement and got back results from query*/

--3.5

```
INSERT INTO BigDataHist (ID, GUID)
VALUES (17, 'Test data input');
```

```
/*Error report -
ORA-00372: file 15 cannot be modified at this time
ORA-01110: data file 15: 'C:\ORACLE19C\ORADATA\ORCL\ARCHIVE\ARCHIVETB01.DBF'*/
```

EXCERCISE 4

--4.1

```
CREATE TEMPORARY TABLESPACE TEMPTB
TEMPFILE 'C:\ORACLE19C\ORADATA\ORCL\TEMP\TEMPTB01.DBF' SIZE 32M
AUTOEXTEND ON;
```

--4.2

```
ALTER DATABASE DEFAULT TEMPORARY TABLESPACE TEMPTB;
```

--4.3

```
SELECT * from DBA_TEMP_FREE_SPACE WHERE tablespace_name='TEMPTB';
```

--4.4

```
select * from BigDataHist;  
select * from bigdata;
```

```
SELECT B.ID, A.GUID  
FROM bigdata A, BigDataHist B  
ORDER BY B.ID;
```

--4.5

I didn't see any difference

EXCERCISE 5

--5.1

--The space used by a table is the space used by all its extents:

```
SELECT SEGMENT_NAME AS Table_Name, TABLESPACE_NAME, BYTES AS Used_Space_Bytes  
, ROUND(BYTES)/1024/1024 AS Used_Space_MB  
FROM USER_SEGMENTS  
WHERE SEGMENT_NAME = 'BIGDATA';
```

--Size is 32 mb or 33554432 bytes

--5.2

```
ALTER TABLE BIGDATA COMPRESS;
```

--5.3

```
SELECT * FROM USED_SPACE
```

```
SELECT TABLESPACE_NAME, SUM(BYTES)/1024/1024 "FREE SPACE(MB)"  
FROM DBA_FREE_SPACE GROUP BY TABLESPACE_NAME;
```

--5.4

```
ALTER TABLESPACE TEMPTB SHRINK SPACE KEEP 16M;
```

--5.5

```

SELECT TF.TABLESPACE_NAME, TF.FILE_NAME , ROUND(TF.BYTES/1024/1024) Total_Size_Mb,
NVL(ROUND(FreeBytes/1024/1024), 0) Free_Space_Mb , NVL(ROUND(UsedBytes/1024/1024), 0)
Used_Space_Mb
FROM DBA_TEMP_FILES TF
LEFT JOIN (
    SELECT File_Id, SUM(BYTES) UsedBytes
    from dba_extents
    group by file_id
) ext on tf.file_id = ext.file_id
LEFT JOIN (
    SELECT File_Id, SUM(bytes) FreeBytes
    from dba_free_space
    group by file_id
) free on tf.file_id = free.file_id

```

```

END;
Error report -
ORA-01653: unable to extend table C##DBADMIN.BIGDATA by 128 in tablespace CURRENTTB
ORA-06512: at line 9
01653. 00000 - "unable to extend table %s.%s by %s in tablespace %s"
*Cause:      Failed to allocate an extent of the required number of blocks for
              a table segment in the tablespace indicated.
*Action:     Use ALTER TABLESPACE ADD DATAFILE statement to add one or more
              files to the tablespace indicated.

```

```

Error starting at line : 156 in command -
INSERT INTO BigDataHist (ID,GUID)
VALUES (17,'Test data input')
Error report -
ORA-00372: file 15 cannot be modified at this time
ORA-01110: data file 15: 'C:\ORACLE19C\ORADATA\ORCL\ARCHIVE\ARCHIVETB01.DBF'

```

TABLESPACE_NAME	FILE_NAME	TOTAL_SIZE_MB	FREE_SPACE_MB	USED_SPACE_MB
1 TEMP	C:\ORACLE19C\ORADATA\ORCL\TEMP01.DBF	129	6	903
2 TEMPTB	C:\ORACLE19C\ORADATA\ORCL\TEMP\TEMP01.DBF	10299	34	30